



Today I am learning to use measurement reasoning to plan, design, investigate and solve problems.

1. Design a Garden Bed

You are designing a rectangular garden bed.

- It must be 2.5 m long.
- It must be 1.2 m wide.
- There will be a path 0.5 m wide around all sides.

a) Draw a plan of the garden bed and the path.



Scale: 1 square = 0.25 m



b) What is the area of the garden bed (not including the path)?

Answer: _____ m²

c) What is the total area including the path?

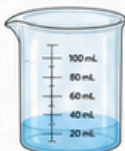
Answer: _____ m²

3. Science Investigation – Water Evaporation

A class investigates how long it takes for water to evaporate.

Each group uses the same amount of water.

Container	Diameter	Water Used	Time to Evaporate
A – Small dish	6 cm	50 mL	2 days
B – Medium dish	12 cm	50 mL	4 days
C – Large dish	18 cm	50 mL	8 days



a) What pattern do you notice about the diameter and the time to evaporate?

b) If Group D uses a dish with a diameter of 24 cm and 50 mL of water, about how long do you think it will take to evaporate? Explain your reasoning.

5. Real-World Measurement Problem

A recipe uses 2.25 L of water.

You only have a measuring jug that holds 750 mL.

a) How many full jugs will you need?

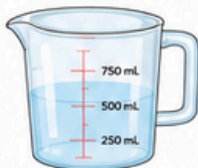
Answer: _____ jugs

b) How much water will be left after the full jugs?

Answer: _____ mL

c) If you then add that remainder once more, what will be the total amount of water?

Answer: _____ L



7. Data and Measurement Reasoning

A student measures the length of a leaf each week.

Week	1	2	3	4	5	6
Length (cm)	2.3	3.1	4.0	4.6	5.2	5.8

a) What is the total increase in length from Week 1 to Week 6?

Answer: _____ cm

b) What is the average increase in length each week?

Answer: _____ cm per week

c) Use the pattern to predict the length in Week 8.

Answer: _____ cm



2. Build a Birdhouse

You are building a birdhouse.

Part	Length (cm)	Width (cm)	Quantity
Front / Back	18	12	2
Sides	18	10	2
Bottom	12	10	1
Roof (each side)	14	12	2



a) What is the total area of wood needed? (Area = length × width)
Show your working.

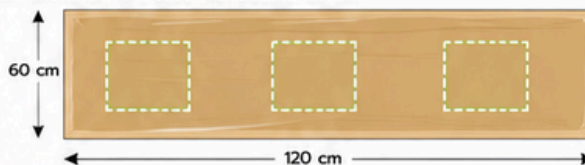
b) If the wood is sold in 500 cm² sheets, how many sheets are needed?

Answer: _____ sheets

4. Convert and Plan

You are planning a school science display table.

- The table space is 120 cm long and 60 cm wide.
- You need to place 3 terrariums on the table.
- Each terrarium is a cube with sides of 20 cm.



a) Will the terrariums fit on the table? Show your measurements.

b) What is the remaining space on the table in centimetres?

Answer: _____ cm length and _____ cm width

6. Engineering Design Challenge

Design a bridge for a small toy car.

- The bridge must be 30 cm long.
- It must be at least 10 cm high.
- It must hold a mass of at least 500 g.
- You have the materials shown.



Available Materials



a) Draw your bridge design.

b) What measurements will you check to test your design?

8. Think and Explain



a) Why is it important to choose the right unit and level of precision when designing or investigating?



b) How does converting measurements help you solve real-world problems?



c) Describe a time when you used measurement in designing, building or investigating something.



I can...

- ☐ plan using appropriate measurements
- ☐ use conversions to solve problems
- ☐ apply measurement in science contexts
- ☐ design and test a solution

Measurement Words



Great job!

You are becoming an amazing problem solver and designer!